

The perpendicular, the oblique line and distance in space

Three lessons on the shortest segment — and the three theorems that compare obliques with it.

LESSON 35–37

3 × 40 MIN

GEOMETRY 10, §15

EXTENSION

LESSON CLOCK

16'

26'

26'

30'

30'

12

Three objects, one picture	16 min
The shortest segment	26 min
Comparing obliques	26 min
Distances	30 min
Practice	30 min
Homework	12 min

LEARNING OBJECTIVES

- Distinguish the perpendicular, an oblique line and its projection.
- Prove that the perpendicular is the shortest segment to a plane.
- Compare obliques by their projections.
- Compute distances from a point to a plane and between parallel planes.

TEXTBOOK

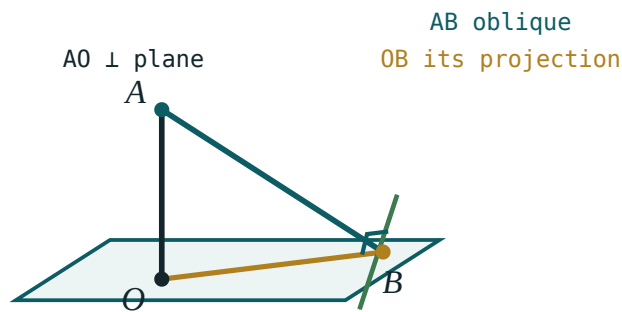
National	pp. 159–176
Cambridge	Core & Extended, pp. 329–334
Workbook	Exercise 15.1

01

Explanation

Three objects, one picture

From a point A not in a plane α , drop the **perpendicular** AO . Any other segment AB from A to the plane is an **oblique**, and OB is its **projection**.



a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

*The perpendicular, the oblique and the projection form
a right triangle in a vertical plane.*

THE RIGHT TRIANGLE

$\triangle AOB$ always has its right angle at O , because $AO \perp \alpha$ and $OB \subset \alpha$. So $AB^2 = AO^2 + OB^2$ — Pythagoras — and every theorem below is a consequence.

The shortest segment

Theorem 1. The perpendicular is shorter than any oblique from the same point.

Proof. $AB^2 = AO^2 + OB^2 > AO^2$ whenever $OB > 0$. ■

DEFINITION OF DISTANCE

The **distance from a point to a plane** is the length of the perpendicular. Theorem 1 is what makes that definition sensible: it is genuinely the shortest.

The same argument, one dimension down, is why the distance from a point to a line in the plane is measured along the perpendicular. Nothing new is being asserted — only extended.

Comparing obliques

Theorem 2. Equal obliques have equal projections, and equal projections give equal obliques.

Theorem 3. Of two obliques, the one with the longer projection is longer.

$$OB_1 > OB_2 \Leftrightarrow AB_1 > AB_2$$

Both follow immediately from $AB^2 = AO^2 + OB^2$: with AO fixed, AB grows exactly as OB does.

A USEFUL COROLLARY

All the obliques of a given length from A land on a **circle** centred at O — because they all have the same projection length. That circle is how a cone meets a plane, and it is the reason a regular pyramid's lateral edges are equal.

The **angle between the oblique and the plane** is $\angle ABO$, the angle with its own projection — and it is the smallest angle AB makes with any line of the plane.

Distances

DISTANCE	HOW IT IS MEASURED
point to plane	the perpendicular from the point
line parallel to a plane, to that plane	from any point of the line — it is constant
between parallel planes	from any point of one to the other
between skew lines	the common perpendicular (Quarter II)

Example. In a cube of edge a , the distance from A to the plane $A_1B_1C_1D_1$ is a ; from A to the plane BDD_1B_1 is half the diagonal AC , that is $\frac{a\sqrt{2}}{2}$.

02 Worked examples

EXAMPLE 1 From a point 10 cm above a plane, two obliques have projections 6 and 8 cm. Find their lengths.

$$1 \quad AB_1 = \sqrt{100 + 36} = \sqrt{136} \approx 11.66$$

$$2 \quad AB_2 = \sqrt{100 + 64} = \sqrt{164} \approx 12.81$$

3 The longer projection gives the longer oblique ✓

ANSWER

≈ 11.7 and ≈ 12.8 cm

EXAMPLE 2 An oblique of length 13 makes 30° with the plane. Find the perpendicular and the projection.

1 The angle is $\angle ABO$.

$$2 \quad AO = 13 \sin 30^\circ = 6.5$$

$$3 \quad OB = 13 \cos 30^\circ \approx 11.26$$

ANSWER

$AO = 6.5$, $OB \approx 11.3$

EXAMPLE 3 Cube of edge 6. Find the distance from A to the plane BDD_1B_1 .

- ① $AC \perp BD$ and $AC \perp BB_1$.
So $AC \perp BDD_1B_1$.

- ② The foot is the centre of the base.

- ③ $AC = 6\sqrt{2}$, half is $3\sqrt{2}$.

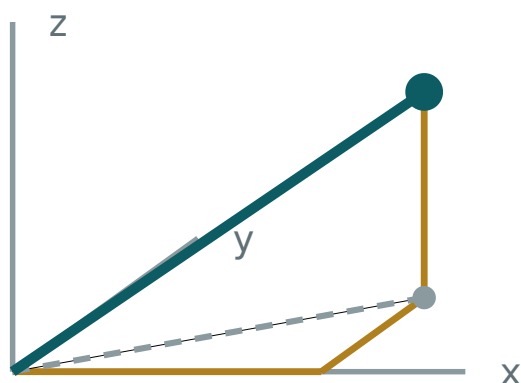
ANSWER

$$3\sqrt{2} \approx 4.24$$

03 **Interactive model**

Hold a stick to the table and rotate it — the shadow lengthens exactly as the stick does.

Perpendicular and oblique



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Perpendicular	Perpendikulyar	Перпендикуляр
Oblique line	Og'ma	Наклонная
Projection of an oblique	Og'ma proyeksiyasi	Проекция наклонной
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Distance from a point to a plane	Nuqtadan tekislikkacha masofa	Расстояние от точки до плоскости
Distance between parallel planes	Parallel tekisliklar orasidagi masofa	Расстояние между плоскостями
Shortest segment	Eng qisqa kesma	Кратчайший отрезок
Equal obliques	Teng og'malar	Равные наклонные
Angle with the plane	Tekislik bilan burchak	Угол с плоскостью

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The distance from a point to a plane is measured along:

any segment

the perpendicular

the oblique

the projection

AB^2 equals:

$AO^2 - OB^2$

$AO^2 + OB^2$

$AO \cdot OB$

$AO + OB$

Equal obliques have:

equal projections

unequal projections

no projection

equal angles only

The longer projection belongs to:

the shorter oblique

the longer oblique

the perpendicular

neither

The angle between an oblique and a plane is measured to:

any line of the plane

its projection

the normal

the edge

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Perpendicular 3, projection 4: the oblique

ANSWER 5

2. Perpendicular 5, projection 12: the oblique

ANSWER 13

3. Oblique 10, perpendicular 6: the projection

ANSWER 8

4. Oblique 25, projection 24: the perpendicular

ANSWER 7

5. Distance from a point to a plane is measured along

ANSWER the perpendicular

6. Equal projections give

ANSWER equal obliques

7. Distance from A to $A_1B_1C_1D_1$ in a cube of edge 6

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. Point 10 above; projections 6 and 8: the obliques

ANSWER $\sqrt{136}$ and $\sqrt{164}$

2. Oblique 13 at 30° : perpendicular and projection

ANSWER 6.5, ≈ 11.3

3. Oblique 20 at 60° : perpendicular

ANSWER $10\sqrt{3} \approx 17.3$

4. Cube edge 6: distance from A to BDD_1B_1

ANSWER $3\sqrt{2}$

5. Perpendicular 8, oblique 17: the projection

ANSWER 15

6. Two obliques 15 and 17 with perpendicular 9: the projections

ANSWER 12 and $\sqrt{208}$

7. Angle of an oblique with perpendicular 6 and projection 8

ANSWER $\approx 36.9^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : distance from A to the plane A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

2. Cube edge 6: distance from B to the plane ACC_1A_1

ANSWER $3\sqrt{2}$

3. A point is 12 from a plane; an oblique makes 40° . Find it

ANSWER ≈ 18.7

4. Three equal obliques from A : what is the foot of the perpendicular?

ANSWER The circumcentre of the three feet

5. A regular pyramid: prove the lateral edges are equal

ANSWER Equal projections — the base circumradius

6. Distance between the parallel planes $ABCD$ and $A_1B_1C_1D_1$ in a cube of edge 9

ANSWER 9

7. Prove that the angle with the projection is the smallest angle with any line of the plane

ANSWER Compare cosines in the right triangles

07 Homework

Homework — 6 tasks

Draw the right triangle AOB separately for every question.

- From a point 8 cm above a plane, two obliques have projections 6 and 15 cm. Find both lengths.
- An oblique of 26 cm makes 30° with a plane. Find the perpendicular and the projection.
- Cube of edge 8: find the distance from A to the plane BDD_1B_1 .
- A perpendicular is 9 and an oblique is 41. Find the projection and the angle with the plane.
- Prove that the perpendicular is shorter than every oblique from the same point.

6. Explain why the lateral edges of a regular pyramid are equal.